

iTutor: A Generative Tutorial System for Teaching the Elders to Use Smartphone Applications

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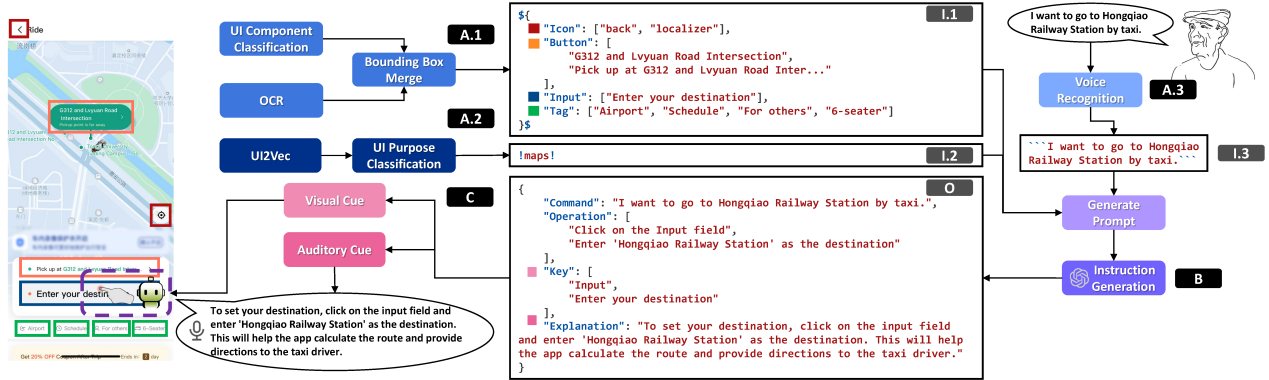


Figure 1: iTutor contains three modules. (A) is the *Metadata Retrieval* module, containing three submodules for extracting the context. (B) is the *Instruction Generation* module, which utilizes large language models to create next-step instructions based on context. (C) is the *Tutorial Generation* module, a natural user interface to provide tutorials for the elders.

ABSTRACT

We present iTutor, a generative tutorial system for promoting smartphone use proficiency among elders. iTutor is unique because it can dynamically generate tutorials based on current operation goals and UI context, which we achieved through leveraging prompt engineering to large language models (LLMs). Our evaluations showed potential for this approach, as we yielded 78.6% accuracy in the instruction generation process. We conclude by providing the roadmap for further development.

CCS CONCEPTS

• **Human-centered computing** → **Accessibility systems and tools.**

KEYWORDS

Accessibility, Tutorial Interface, Prompt Engineering

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1 INTRODUCTION

As digitization occurs in almost every walk of life, we can now get most services straight from smartphone applications. However, accessibility to this convenience is not equal for everyone. Research has found that digital accessibility tends to be stratified by age, a phenomenon known as the digital divide [13].

One way of bridging this accessibility gap is by introducing better tutorial interfaces that are tailor-made for elders. For example, Synapse [7] offers elders multimodal, step-by-step instructions for using multiple apps. Besides, it also supports the trial-and-error learning mode, a proven method for improving source memory [4]. Although systems of this type can be good tutors for elders, they need more versatility. In the case of Synapse, all the instructions are recorded in advance, limiting the use cases for the system when faced with unlisted scenarios.

Another way of bridging this accessibility gap is through automatic execution systems. Prompted by natural language commands, these systems parse the user's command and execute on behalf of them across single [14] or multiple [12] UI pages. Yet the amenity of this type of system can not address all the challenges for elders in using smartphone applications. For example, if the system glitches, it may lead elders to abandon the current task altogether [1].

In this poster, we propose iTutor, a generative tutorial system for teaching the elders to use smartphone applications by providing in-context continuing support. iTutor combines the merits while addressing the limitations of the previous two approaches through

dynamic tutorial generation for each UI page. It enables the elders to follow a step-by-step tutorial while obtaining a correctable, semi-automatic execution process.

2 SYSTEM

The pipeline of iTutor includes three modules (Fig.1): *Metadata Retrieval*, *Instruction Generation*, and *Tutorial Interface Generation*.

2.1 Metadata Retrieval

Metadata retrieval (Fig.1(A)) is the first step in iTutor's dataflow. The goal for retrieving metadata is to transform other modalities into text, the only acceptable input for most LLMs. The image of the current UI page and the user's voice input are the two input modalities that require conversion.

On converting the image of the current UI page to text, iTutor focuses on two types of UI information: UI components (Fig.1(A.1)) and the general purpose of the UI page (Fig.1(A.2)). Earlier research has proposed traditional [2] and modern [16] approaches for extracting UI components, which iTutor intends to reference. Meanwhile, we are in the process of collecting a new dataset to reflect the latest UI design styles [15]. For analyzing the general purpose of the UI page, we propose formulating it as a downstream task of the UI2Vec model¹. We obtained 95% accuracy in classifying the Enrico [8] dataset's 20 classes.

iTutor uses voice recognition to convert the user's voice input to text (Fig.1(A.3)). However, we do not include a separate module for this task, as it is readily available through mobile OS's APIs.

The three types of information retrieved from the two input modalities are the context we subsequently use to prompt the LLM. We organize the voice input content and the general purpose of the UI page in text (Fig.1(I.2, I.3)) and store the UI element's metadata in JSON format (Fig.1(I.1)).

2.2 Instruction Generation by LLMs

Instruction generation (Fig.1(B)) is the core of the iTutor system, and we propose using large language models (LLMs) to fulfill this task. Research has found that language models can serve as a knowledge base [9] while they learn from few-shot examples [3]. Thus, we elicit its UI knowledge through carefully designed prompts.

The present iteration of iTutor utilizes chatGPT² to generate instructions, with specific prompting techniques [5, 11] applied to enhance its zero-shot performance.

Our prompt consists of three parts: 1) we inform chatGPT to provide instructions for navigating through a given application, 2) we provide chatGPT with the three types of UI context with specific markings, and 3) we request chatGPT to respond in JSON including the original command, next-step instruction, UI element to operate on, and the explanation (Fig.1(O)). We embed a cloze-formed prompt template into iTutor, where iTutor can automatically fill in the UI context, requiring no further user action.

We then evaluated the prompt with a small dataset that included three popular apps with manually constructed UI context³. We use the UI contexts to prompt chatGPT's user interface, refreshing the

context for every input. Results indicated that chatGPT can provide the correct next-step operation in 78.6% (22/28) of the cases.

However, our current approach is not without limitations. Firstly, chatGPT is prone to fail when presented with icons. Icons are ambiguous in conveying the aim of UI elements. For example, two failed cases were caused by using the 'upload' icon as the sharing icon. Secondly, the JSON data we used to prompt chatGPT can only present what is visible on the UI page. Therefore, when faced with operations like swiping, chatGPT might produce wrong results. In addition, we observed that due to the lack of UI state in the JSON data, chatGPT might repeat the previous step and get the intended operation in the second step, a correct yet undesired response.

Additionally, we recognize that the instructions provided in the JSON response may still be unclear for elders to comprehend. To address this problem, we use the *Tutorial Interface Generation* module to transcribe the JSON file to an interactive interface.

2.3 Tutorial Interface Generation

Tutorial Interface Generation (Fig.1(C)) is where iTutor transcribes the JSON response into an interactive tutorial. Based on the feedback from cognitive walkthroughs with elders, we prototyped a Natural User Interface (NUI) to align with the cognition patterns of the elders.

We outline three considerations in designing this NUI: 1) convey instructions with voice via a voice user interface (VUI) to alleviate cognitive burdens in using a graphical user interface (GUI) [10], 2) display the intended operation with highlighted visual cues, 3) generate dynamic, in-context tutorials with every change on the UI interface. Overall, iTutor provides continuing support to elderly users via an immersive NUI with visual and auditory interactions.

We plan to conduct a user study once iTutor is fully optimized. We will recruit participants between 60 and 75 unfamiliar with smartphone usage. We will instruct the participants to perform the following tasks using iTutor: making taxi calls, using navigation, mobile payments, and phone calls, and quantitatively evaluate the design with an eye-tracking test [6]. We will then revise iTutor based on this feedback.

3 CONCLUSION

With its ability to generate interactive tutorials based on UI context, iTutor presents a new solution to bridge the digital divide. Early experiments on the UI2Vec model and the Instruction Generation module have shown promising prospects for our design. The adaptability of iTutor on different systems and applications should provide it with high compatibility with new and existing devices.

In the future, other than completing the implementation of the system, we hope to design and evaluate different interaction designs to maximize the performance of iTutor. Besides that, it will also be meaningful if iTutor can run offline while maintaining the instruction generation performance, which will allow iTutor to cover more use case scenarios.

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¹<https://github.com/Motion115/UI2Vec>

²<https://chat.openai.com/>

³<https://motion115.github.io/iTutor>

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